# Aim

To write a program to find the 1-norm, 2-norm and infinity norm of the matrix and display the result in two decimal places.

# Equipment's required:

1. Hardware – PCs
2. Anaconda – Python 3.7 Installation / Moodle-Code Runner

# Algorithm:

```
1. Get the input matrix using np.array()
2. Find the 2-norm of the matrix using np.linalg.norm()
3. Print the norm of the matrix in two decimal places.
```

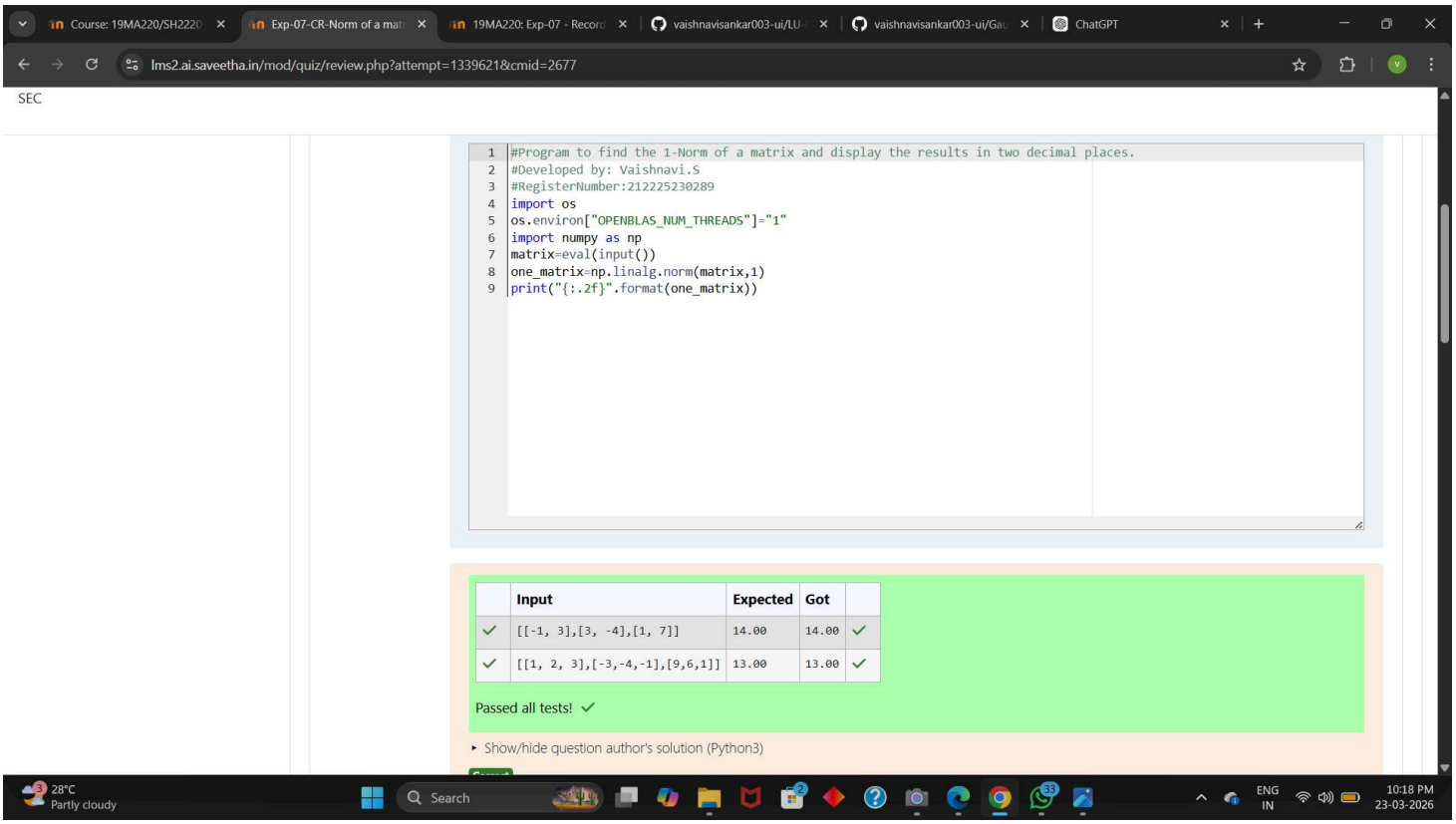
# Program:

```
# Register No:212225230289
# Developed By:vaishnavi s
# 1-Norm of a Matrix
import os
os.environ["OPENBLAS_NUM_THREADS"]="1"
import numpy as np
matrix=eval(input())
one_matrix=np.linalg.norm(matrix,1)
print("{:.2f}".format(one_matrix))



# 2-Norm of a Matrix

import os
os.environ["OPENBLAS_NUM_THREADS"]="1"
import numpy as np
matrix=eval(input())
two_matrix=np.linalg.norm(matrix,2)
print("{:.2f}".format(two_matrix))


# Infinity Norm of a Matrix
```

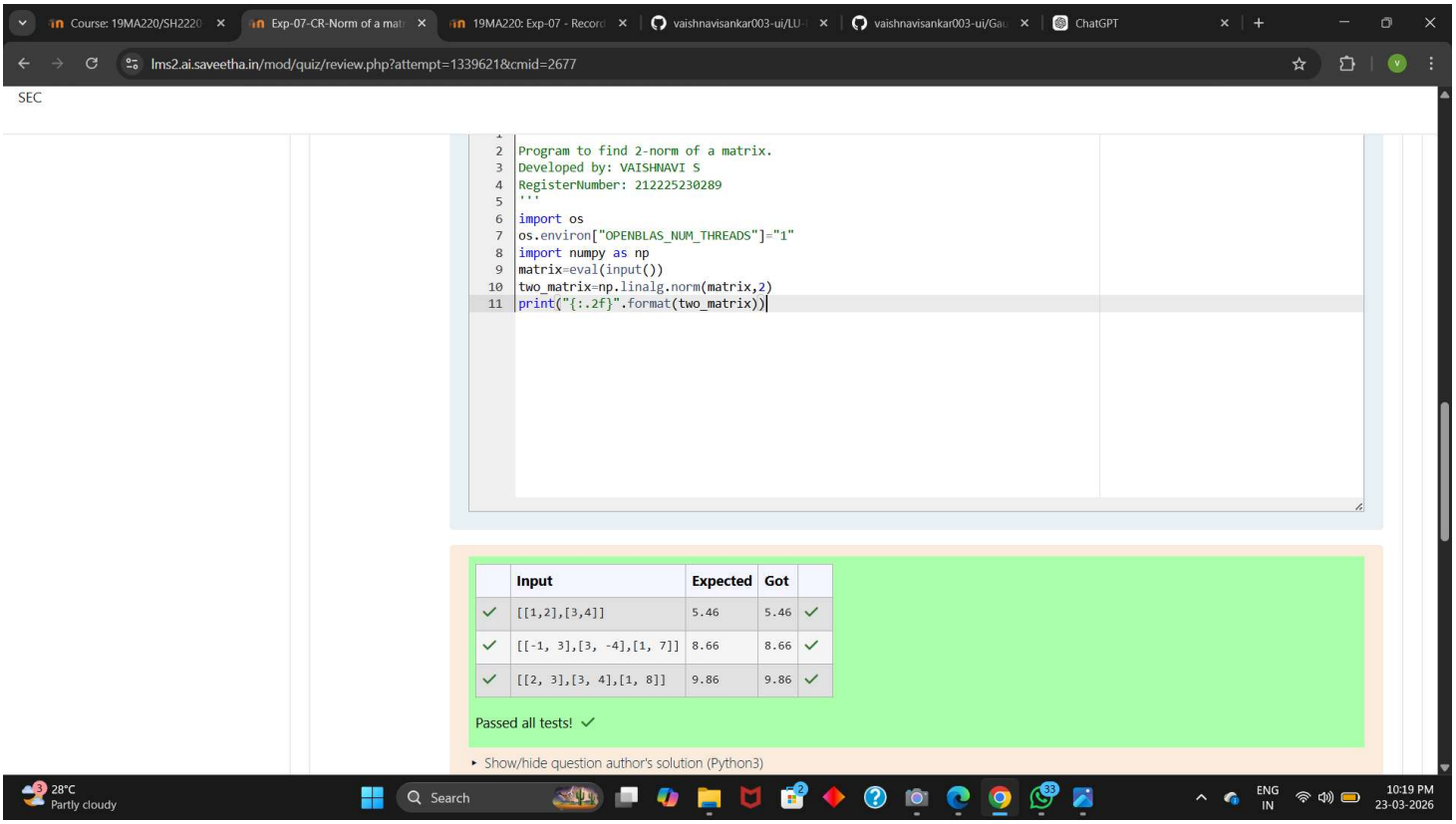
```
import os
os.environ["OPENBLAS_NUM_THREADS"]="1"
import numpy as np
matrix=eval(input())
inf_matrix=np.linalg.norm(matrix,np.inf)
print("{:.2f}".format(inf_matrix))
```

# Output:

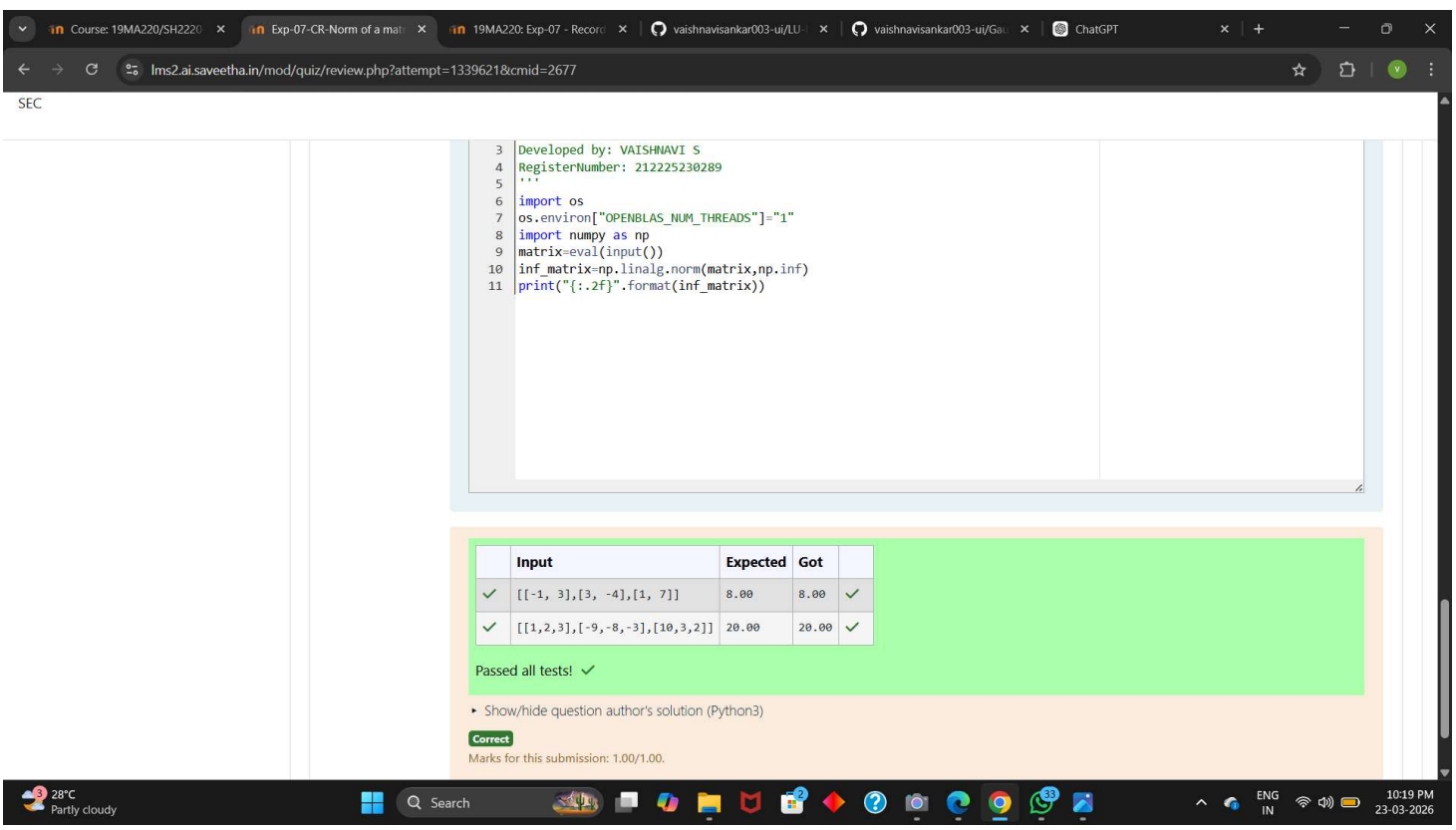
## 1-Norm of a Matrix




## 2-Norm of a Matrix




## Infinity Norm of a Matrix




# Result

Thus the program for 1-norm, 2-norm and Infinity norm of a matrix are written and verified.